Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

1. Which of the following are local explanations for LLMs (check all that apply)? 1 / 1 point

- [x] Feature attribution

  Correct
  Correct! Feature attribution is a local explanation approach used in LLMs.

- [ ] Concept-based explanation
- [x] Example-based explanation

  Correct
  Correct! Example-based explanation is a local explanation approach used in LLMs.

2. Fill in the blank: Neural models are highly vulnerable to carefully crafted, small modifications in the input data that can drastically alter the model's predictions, despite being nearly imperceptible to humans. These are called Adversarial examples. 1 / 1 point

Response 1

- ( ) Influential
- (•) Adversarial
- ( ) Counterfactual

  Correct
  Correct! Adversarial examples are carefully crafted, small modifications in the input data that can drastically alter the model's predictions, despite being nearly imperceptible to humans.

3. Which type of probing-based explanation examines coreference, entities, and relations? 1 / 1 point

- (•) Semantic
- ( ) Parameter-free
- ( ) Syntax

  Correct
  Correct! Semantic probing examines coreference, entities, and relations.

4. The lack of what makes evaluating component-level explanations challenging? 1 / 1 point

- (•) Ground truth annotations
- ( ) Transparency
- ( ) Computational resources

  Correct
  Correct! The lack of ground-truth annotations makes evaluating component-level explanations challenging.

5. Which of the following XAI methods was NOT used for explaining how CoT prompting affects the behavior of LLMs? 1 / 1 point

- ( ) Saliency Scores
- ( ) Counterfactual Prompts
- (•) Adversarial Examples

  Correct
  Correct! In the lecture, we did not discuss adversarial examples being used to explain how CoT prompting affects the behavior of LLMs.

6. If you want to handle complex, nonlinear relationships in embeddings, what dimensionality reduction / embedding visualization technique should you use? 1 / 1 point

- ( ) t-SNE
- ( ) PCA
- (•) UMAP

  Correct
  Correct! UMAP is the best method if you need to ensure you can handle complex, nonlinear relationships in embeddings.

7. What did Bau et.al. combine with a traditional network dissection XAI approach when applying the method to GANs? 1 / 1 point

- ( ) A counterfactual explanation framework
- (•) A segmentation network
- ( ) A multilayer perceptron

  Correct
  Correct! By combining network dissection with a segmentation network, Bau, et.al. were able to answer a number of interesting questions.

8. What were methods for Latent Space Guided Perturbation in GANs discussed in the lecture (check all that apply)? 1 / 1 point

- [x] Support Vector Machine (SVM)

  Correct
  Correct! Shen, et.al. 2020 trained linear support vector machines (SVM) to find the separation hyperplanes/subspaces in the latent space for latent space guided perturbation in GANs.

- [x] Principal Component Analysis (PCA)

  Correct
  Correct! Härkönen, et.al. 2020 applied PCA in GAN latent space and found it gives semantically significant directions.

- [ ] LIME

9. What is the name of the framework to produce pixel-level attribution maps by upscaling and aggregating cross-attention word-pixel scores in the denoising subnetwork? 1 / 1 point

- (•) DAAM
- ( ) LAAM
- ( ) BAAM

  Correct
  Correct! In 2022, Tang, et.al. proposed "DAAM", a Cross-Attention Pixel-Attribution Visualization approach.

10. Which of the following were local explanations used by Nagisetty, et.al. in xAI-GAN (check all that apply)? 1 / 1 point

- [x] LIME

  Correct
  Correct! LIME was one of several local explanation methods that was explored by Nagisetty for xAI-GAN.

- [ ] TransSHAP
- [x] DeepSHAP

  Correct
  Correct! DeepSHAP was one of several local explanation methods that was explored by Nagisetty for xAI-GAN.

- [x] Saliency Maps

  Correct
  Incorrect. Correct! Saliency Maps were one of several local explanation methods that was explored by Nagisetty for xAI-GAN.